Arc synthesis: the gamma rung

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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Abstract

Seven tides carried the seabed's anharmonic machinery from first to second order and cashed the result as a recovery theorem: for the potential $\frac{\lambda}{2}x^2+\frac{\alpha}{6}x^3+\frac{\gamma}{24}x^4$, the fourth cumulant of the Gibbs measure satisfies $t^3\kappa_4 \to 3\alpha^2/\lambda^5-\gamma/\lambda^4$, and consequently the complete jet $(\lambda, \alpha, \gamma)$ is determined by observable data — the germbij note's Theorem 3.1 triangular induction, formalised through its third rung. The arc is a case study in programme-level scoping: the target was pinned numerically and its cancellation table verified symbolically before any Lean, and neither moved in seven tides.

1 The mathematics

The germbij note's regular case (Theorem 3.1) recovers the Taylor coefficients of a loss at a non-degenerate minimum from susceptibilities, one order at a time. The seabed's Stage-2 programme (May 2026) had the first two rungs' inputs: leading asymptotics for the mean and covariance susceptibilities, from which (λ, α) recovery was read off earlier in the recovery thread. The third rung needs γ, which first appears in the *fourth cumulant* at order t^{-3} — behind an exact cancellation of the Gaussian t^{-2} terms, so the leading-order machinery cannot see it. The programme therefore had to deepen the entire analytic stack by one order:

Stage 1 — the exponential Taylor remainder for signed arguments, $|e^{-s} - \sum_{j<n}(-s)^j/j!| \leq \frac{|s|^n}{n!}\max(1, e^{-s})$, because the rescaled perturbation $s_t(u) = Au^3/\sqrt{t} + Bu^4/t$ changes sign.

Stage 2 — the cubic-order integral remainder ($O(1/(t\sqrt{t}))$, reusing the seabed's two-branch Gaussian decay verbatim) and the quadratised J_n expansion $J_n = m_n - \frac{A}{\sqrt{t}}m_{n+3} - \frac{B}{t}m_{n+4} + \frac{A^2}{2t}m_{n+6} + O(1/(t\sqrt{t}))$, whose A^2 coefficient is the payload.

Stage 3 — second-order rates for the Gibbs moments themselves, each riding an exact $1/t$ cancellation in $J_r - (\text{leading} + \text{coeff}/t)J_0$ verified by `ring` on explicit coefficients.

Stages 4–5 — the limit: $t^3\kappa_4$ splits by one `ring` identity into five rate-convertible pieces; the constants sum to $108A^2 - 24B$ over λ^2, which is $3\alpha^2/\lambda^5 - \gamma/\lambda^4$.

Stage 6 — the recovery: composing with the merged (λ, α) recovery, uniqueness of limits pins γ. Fifteen lines, first-attempt clean.

2 What made it work

Scope before stage 1. The scoping consult received the seabed inventory and a numerically pinned target (high-precision quadrature with a spike breakpoint — the adaptive integrator misses the $O(t^{-1/2})$-width peak without one), verified the full symbolic route including the fourth-cumulant cancellation table, fixed a six-stage plan, and named the risks. Both named risks were already discharged: the coercivity risk by the seabed's standing discriminant hypothesis, the domination risk by the existing `ScalarBound` layer. Nothing in the plan was revised during execution.

The catalogue compounds. Across seven tides, every build failure belonged to a documented class — and twice the failure was one this very programme had documented days earlier (the radical-atom rewrite, the type-ascribed constant witness). New entries added: the $\sqrt{t}$-atom substitution for mixed $t/\sqrt{t}$ identities, the `Pi.sub` member of the `Pi`-form family, `set_option` placement relative to docstrings, and `gcongr`'s self-sufficiency with context hypotheses. Two tides built first-try, including the finale.

Checkpointed WIP commits. The two largest files (the moment rates at ∼700 lines, stage 2's layers) were built layer by layer with building, sorry-free commits at each seam, so no session boundary or error ever implicated more than the newest layer.

Exact algebra over estimates. At every level — the J_n terms, the moment ratios, the cumulant combination — the design keeps closed forms exact until a single explicitly bounded remainder, so `ring` carries the mathematical content and the analysis reduces to one squeeze per level.

3 What it cost

Seven tides (stage 2 split at its natural seam), roughly 1800 lines of new Lean, five consults (one programme-scoping, the rest inherited), and two numerical checks executed before their Lean. The de-privatised seabed lemmas (`integrable_J_n`, `J_0_eventually_bounded`) and the promoted catalogue entries are the reusable residue.

4 What remains

The rungs above γ (fifth cumulant onward) are structurally identical ladders over the same machinery, one Taylor order deeper per rung; a general-order Theorem 3.1 would want the pattern abstracted rather than iterated. The multivariate regular case and the note's remaining programmes (Proposition 4.1's flat counterexample, multivariate Section 7.4) are separate commissions.